



Unseen Model-Based Pose Estimation and Hardware-in-the-Loop Testing for Uncooperative Spacecraft with Monocular Vision

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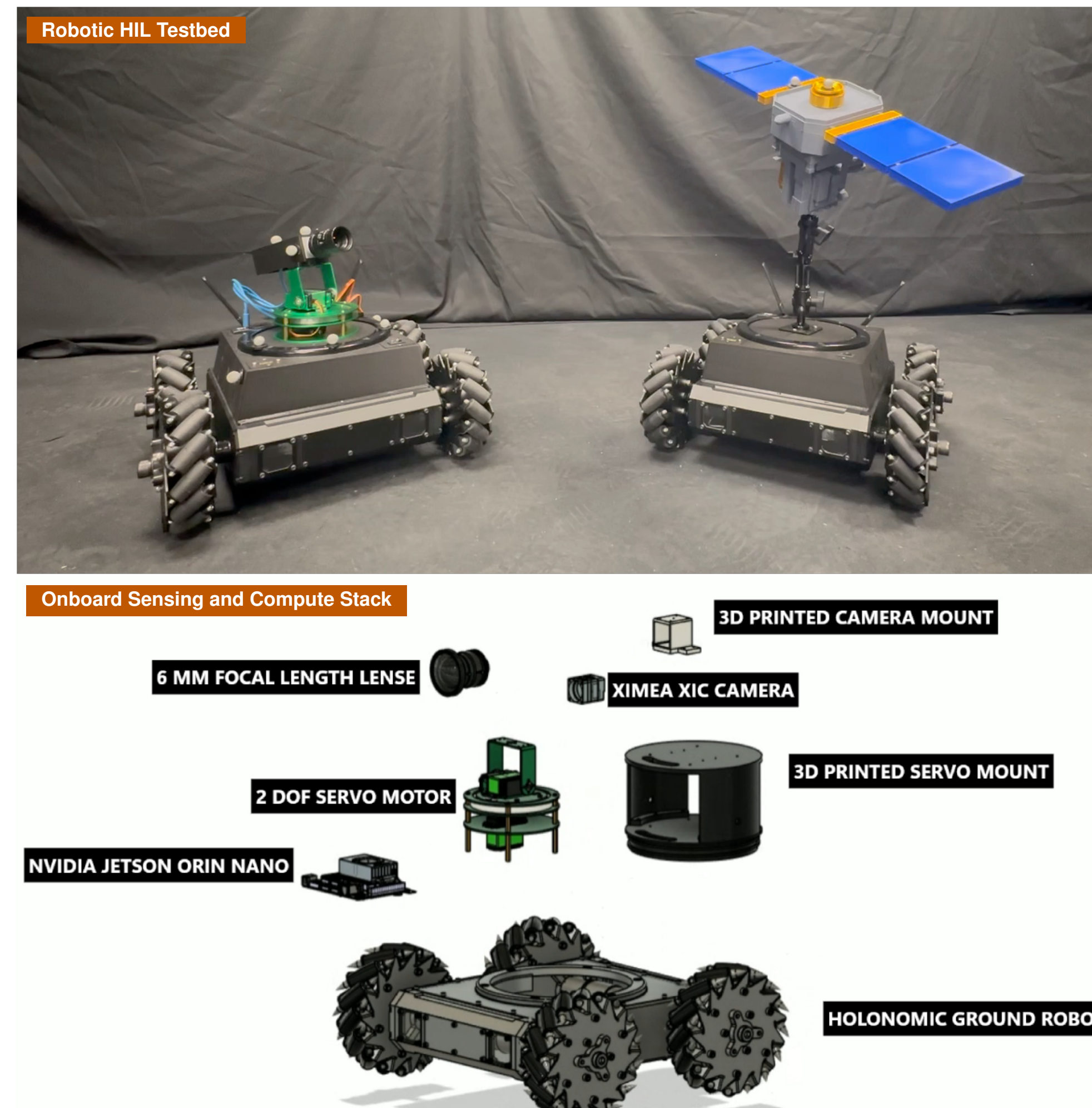
Abstract

Monocular relative pose estimation supports autonomous **Rendezvous, Proximity Operations, and Docking (RPOD)** and **On-Orbit Servicing, Assembly, and Manufacturing (OSAM)** for non-cooperative spacecraft. We demonstrate an unseen model-based recovery and tracking pipeline that uses a monocular **Red-Green-Blue (RGB)** image and target **Computer-Aided Design (CAD)** model rather than target-specific pose-training imagery. The system runs on a robotic **Hardware-in-the-Loop (HIL)** testbed with motion-capture-based truth correction and onboard compute.

Motivation and Problem Setup

- Non-cooperative targets may not transmit relative state or provide fiducials; the chaser must infer pose onboard.
- Target-specific pose networks require labeled or synthetic imagery and retraining for each known target.
- This model-based pipeline assumes a target three-dimensional (3D) model and avoids target-specific pose-training images.
- HIL testing exposes sim-to-lab appearance, compute limits, truth collection, and mockup fidelity.

Experimental Platform and Truth Data



Two holonomic robots emulate chaser-target motion; the lower view shows the onboard sensing and compute stack.

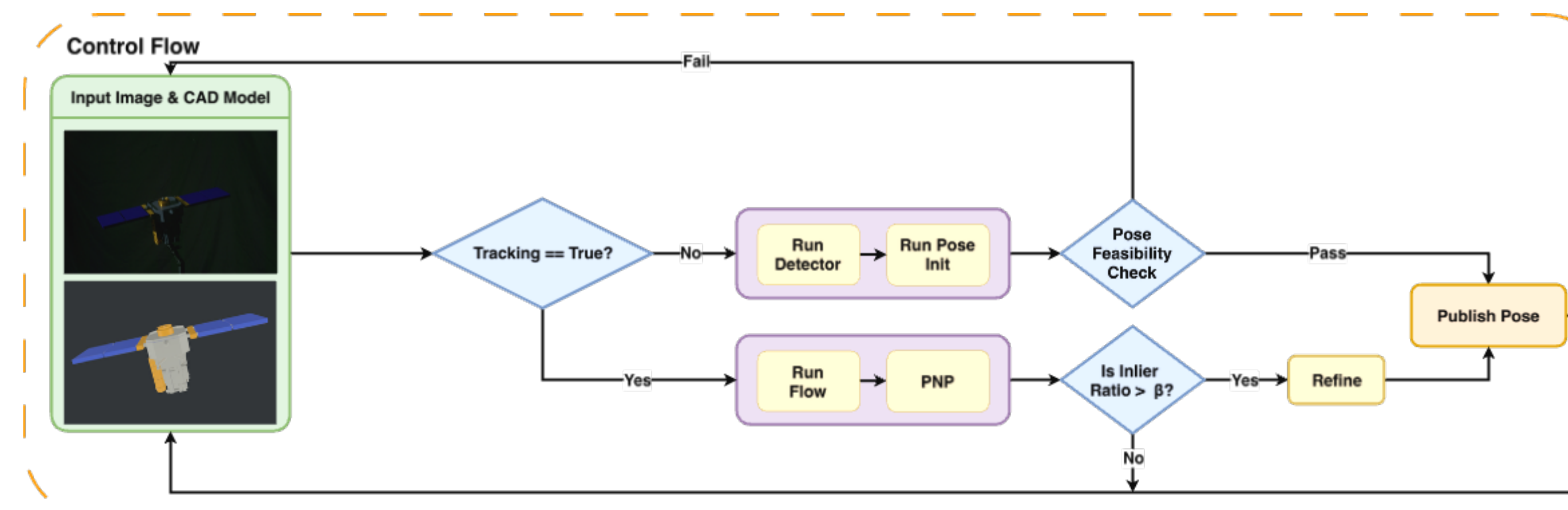
Demonstration Stack

- Camera: Basler Ace (acA4112-30uc); compute: NVIDIA Jetson Orin NX 16 GB.
- Emulation uses Distance Unit (DU) ratio 1:20 and Time Unit (TU) ratio 10:1.
- Robot Operating System 2 (ROS 2) nodes run the measurement model/filter and collect imagery plus Vicon data.

Truth-Like Poses

- Vicon motion capture supplies truth-like camera-target poses, but its frame definitions introduce offsets.
- Open Source Computer Vision Library (OpenCV) calibration estimates camera-to-board correction terms.
- Residual error can arise from calibration noise, Vicon coverage limits, and print/CAD mismatch.

Method Overview



The measurement model alternates between recovery and frame-to-frame tracking; feasibility and inlier checks determine publication or reacquisition.

Inputs

- Monocular RGB image, target CAD/mesh model, camera calibration, and filter prediction.

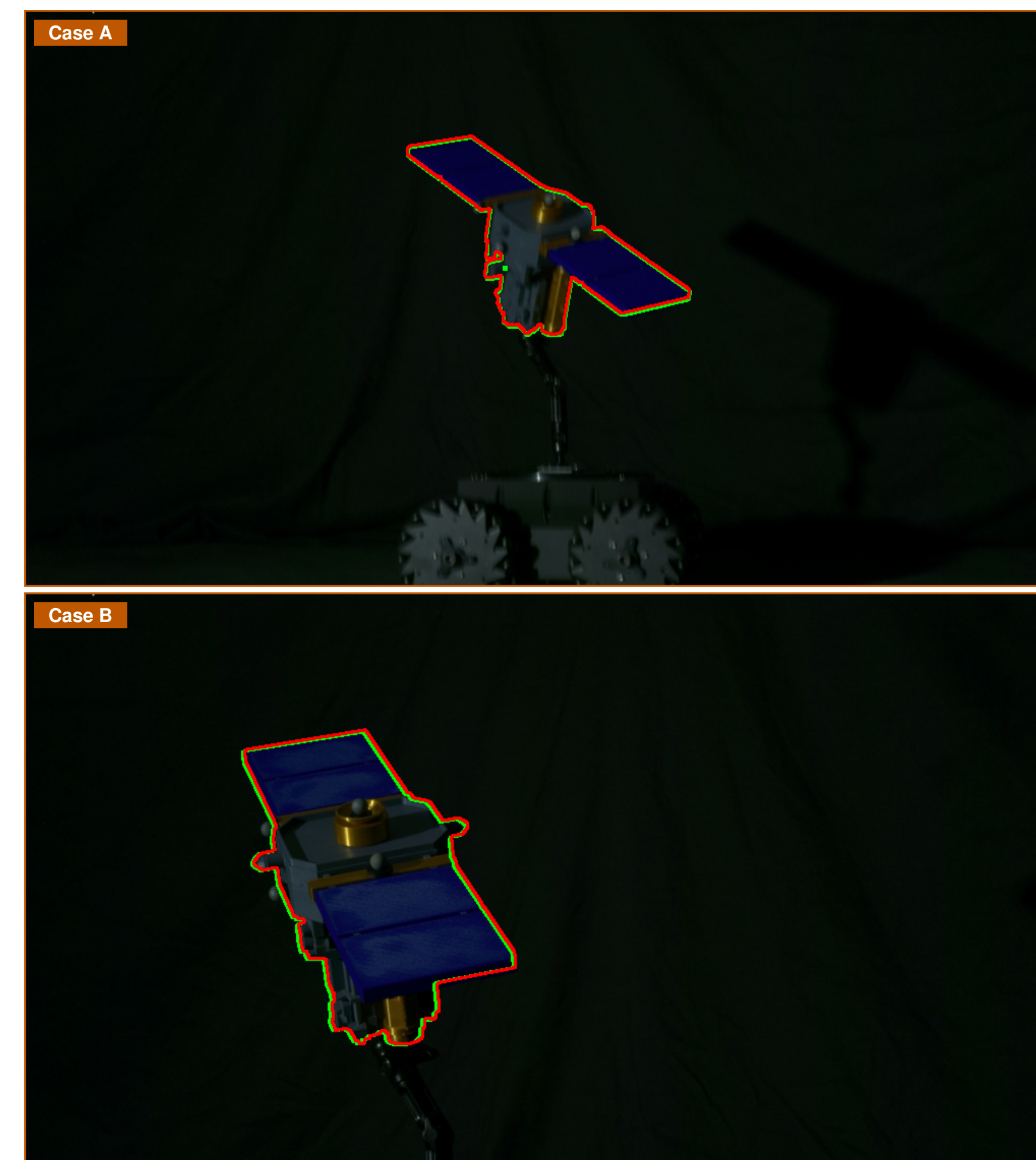
Recovery Mode

- Localize the target and initialize pose with model-based recovery.
- Check detector support against the projected model and seed image-to-model correspondences.

Tracking Mode

- Propagate correspondences with optical flow.
- Run **Perspective-n-Point (PnP)**, refine, monitor inliers, and fall back to recovery if tracking degrades.

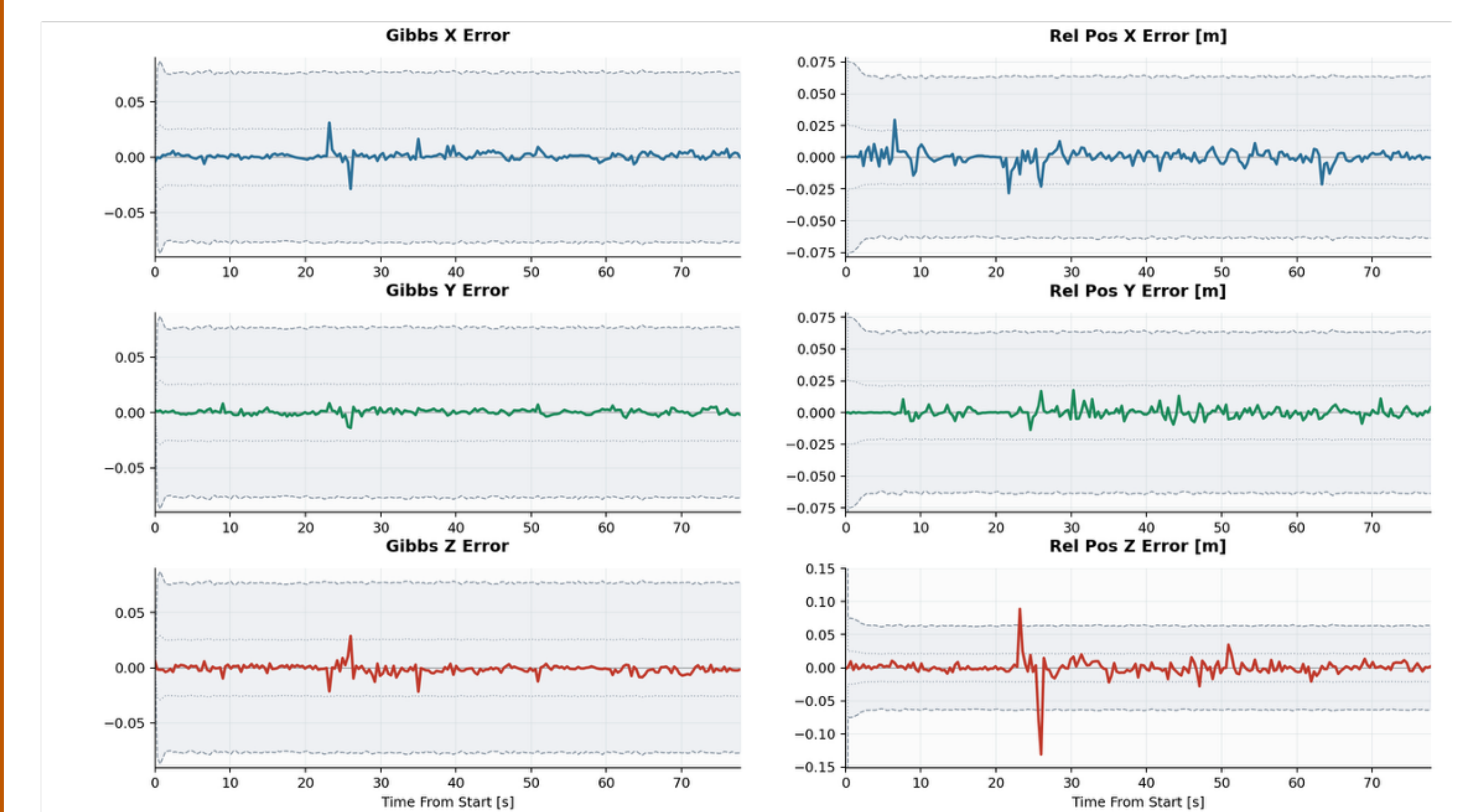
Qualitative Results



What the Overlays Check

- Green contours show measurement-model output; red contours show the filter state after the measurement update.
- Visual agreement indicates that projected poses are plausible for these frames.
- Differences between contours and error spikes are treated as diagnostics for timing, noise, and filter tuning.

Quantitative Results



Performance average	Value
Translation, in-plane Root Mean Square Error (RMSE)	0.48 cm
Translation, depth RMSE	1.68 cm
Euler rotation RMSE	1.05 deg
Measurement frequency	2.85 Hz

Interpretation

- Error plots report attitude and translation error for a single demonstration run.
- The table preserves the reported in-plane, depth, attitude, and measurement-frequency averages.
- Additional tuning and parameter testing are needed to further validate consistency.

Conclusions and Future Work

Takeaways

- Demonstrated an unseen model-based monocular pose pipeline on robotic HIL hardware with onboard processing and filter integration.
- Recovery plus tracking combines global reinitialization with optical-flow correspondence propagation and PnP/refinement.
- The experiment directly exercises truth acquisition, sim-to-lab constraints, and edge-compute throughput.

Next Steps

- Validate over many runs and targets.
- Tune timing, data-rate synchronization, and filter covariance for smoother updates.
- Improve rendering/refinement speed with template warping.
- Test in another facility and transition modular ROS 2 nodes to other groups.

Scope: results are limited to this Year 2 HIL test article and trajectory; broader runs, targets, and facilities remain future validation. Supported by the United States Space Force (USSF) Space Strategic Technical Institute for In-Space Operations, FA9453-24-2-0004.